

Table S1. Parameter explanations for the PM<sub>2.5</sub> prediction model. This table lists the input parameters used in the PM<sub>2.5</sub> concentration prediction model and provides a brief explanation of their impact on air quality.

Table S2. Air quality level classification based on the "Technical Regulations for Ambient Air Quality Index".

Table S3. Comparative experimental results of different methods in the BTH region.

Table S4. Comparative experimental results of different methods in the CSJ region.

Fig.S1. Sensitivity analysis of different terrain constraint thresholds on the validation set for 24-hour forecasting.

Fig.S2. Distribution of stations in the BTH Region.

Fig.S3. Comparison of different spatial regularization strategies.

Table S1. Parameter explanations for PM<sub>2.5</sub> prediction model.

| Parameter                                  | Explanation                                                                                                                                                                                                                                                                           |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AQI (Air Quality Index)                    | A single numerical indicator that integrates multiple pollutant concentrations (e.g., PM <sub>2.5</sub> , PM <sub>10</sub> , O <sub>3</sub> ) to represent overall air quality. It helps track general pollution levels and provides indirect insights into PM <sub>2.5</sub> trends. |
| CO (Carbon Monoxide)                       | Concentration of Carbon Monoxide, mainly from vehicle exhaust and combustion sources. CO often co-occurs with PM <sub>2.5</sub> in urban areas, indicating shared emission origins.                                                                                                   |
| NO <sub>2</sub> (Nitrogen Dioxide)         | A reactive nitrogen gas produced mainly by fossil fuel combustion. It is a precursor to secondary particulate formation and correlates with traffic-related PM <sub>2.5</sub> emissions.                                                                                              |
| O <sub>3</sub> (Ozone)                     | A secondary pollutant formed through photochemical reactions involving NO <sub>x</sub> and VOCs. Ozone levels influence the oxidative environment and the formation of secondary PM <sub>2.5</sub> .                                                                                  |
| PM <sub>10</sub> (Particulate Matter 10)   | Particulate matter with diameter $\leq 10\mu\text{m}$ , which includes PM <sub>2.5</sub> . PM <sub>10</sub> trends often move with PM <sub>2.5</sub> and can indicate broader particle pollution.                                                                                     |
| PM <sub>2.5</sub> (Particulate Matter 2.5) | Fine particulate matter with diameter $\leq 2.5\mu\text{m}$ . It poses significant health risks and is the core target variable in the prediction model.                                                                                                                              |
| SO <sub>2</sub> (Sulfur Dioxide)           | A gaseous pollutant from industrial processes and coal combustion. SO <sub>2</sub> can react in the atmosphere to form sulfate particles, contributing to PM <sub>2.5</sub> levels.                                                                                                   |
| t2m (2-meter Temperature)                  | Near-surface air temperature at 2 meters. It influences atmospheric mixing, stability, and chemical reaction rates, all of which affect PM <sub>2.5</sub> dynamics.                                                                                                                   |
| d2m (2-meter Dew Point Temperature)        | Dew point temperature at 2 meters, reflecting ambient humidity. High dew points are associated with enhanced aerosol hygroscopic growth and secondary PM formation.                                                                                                                   |
| V10 (10-metre v-component of wind)         | Meridional (north-south) wind component at 10 meters altitude. It affects pollutant transport and regional dispersion of PM <sub>2.5</sub> .                                                                                                                                          |
| u10 (10-metre u-component of wind)         | Zonal (east-west) wind component at 10 meters altitude. Together with V10, it determines wind-driven horizontal movement of air pollutants.                                                                                                                                           |
| tp (Total Precipitation)                   | Total precipitation rate in millimeters per hour. Rainfall helps remove PM <sub>2.5</sub> from the air through wet deposition.                                                                                                                                                        |
| sp (Surface Pressure)                      | Surface atmospheric pressure, indicating synoptic weather conditions. It influences boundary layer height and the vertical mixing of pollutants, thus affecting PM <sub>2.5</sub> accumulation.                                                                                       |

Table S2. Air quality level classification based on the "Technical Regulations for Ambient Air Quality Index"

| Level              | PM2.5 concentration range (µg/m³) |
|--------------------|-----------------------------------|
| Excellent          | 0-35                              |
| Good               | 36-75                             |
| Light Pollution    | 76-115                            |
| Moderate Pollution | 116-150                           |
| Heavy Pollution    | 151-250                           |
| Severe Pollution   | >250                              |

Table S3.Comparative experimental results of different methods in the BTH region.

| Model                | Metric | +1h         | +6h          | +12h         | +24h          |
|----------------------|--------|-------------|--------------|--------------|---------------|
| Persistence          | MAE    | 4.907       | 9.708        | 13.245       | 17.088        |
|                      | RMSE   | 8.126       | 16.521       | 21.998       | 27.371        |
|                      | R2     | 0.941       | 0.756        | 0.568        | 0.332         |
| Seasonal Persistence | MAE    | 5.679       | 10.269       | 13.604       | 17.012        |
|                      | RMSE   | 9.452       | 17.227       | 22.379       | 27.250        |
|                      | R2     | 0.920       | 0.734        | 0.552        | 0.334         |
| ARIMA                | MAE    | 4.906       | 9.618        | 13.241       | 17.081        |
|                      | RMSE   | 8.115       | 16.539       | 22.010       | 27.360        |
|                      | R2     | 0.942       | 0.757        | 0.568        | 0.333         |
| GRU                  | MAE    | 5.354±0.083 | 10.453±0.092 | 12.651±0.118 | 14.068±0.220  |
|                      | RMSE   | 8.521±0.126 | 15.446±0.155 | 19.362±0.174 | 21.354±0.310  |
|                      | R2     | 0.932±0.012 | 0.762±0.006  | 0.665±0.019  | 0.601±0.025   |
| STGCN                | MAE    | 5.681±0.061 | 8.856±0.175  | 11.543±0.125 | 14.477± 0.325 |
|                      | RMSE   | 8.428±0.095 | 13.787±0.123 | 17.708±0.159 | 21.618±0.354  |
|                      | R2     | 0.930±0.014 | 0.829±0.010  | 0.719±0.012  | 0.630±0.016   |
| T-GCN                | MAE    | 4.979±0.054 | 8.858±0.119  | 11.560±0.215 | 13.903±0.251  |
|                      | RMSE   | 7.585±0.090 | 13.779±0.156 | 17.645±0.156 | 20.529±0.289  |
|                      | R2     | 0.940±0.005 | 0.837±0.004  | 0.750±0.005  | 0.641±0.009   |
| MTGNN                | MAE    | 5.021±0.061 | 9.199±0.243  | 11.359±0.254 | 15.120±0.432  |
|                      | RMSE   | 7.713±0.100 | 14.591±0.154 | 17.109±0.359 | 23.508±0.483  |
|                      | R2     | 0.935±0.011 | 0.832±0.009  | 0.698±0.030  | 0.520±0.029   |
| HighAir              | MAE    | 4.899±0.050 | 8.852±0.135  | 11.552±0.211 | 13.856±0.158  |
|                      | RMSE   | 7.603±0.095 | 13.720±0.120 | 16.896±0.151 | 20.403±0.215  |
|                      | R2     | 0.941±0.003 | 0.838 ±0.007 | 0.750 ±0.008 | 0.642±0.011   |
| GAGNN                | MAE    | 4.942±0.055 | 9.046±0.251  | 11.477±0.100 | 14.007±0.219  |
|                      | RMSE   | 7.402±0.090 | 13.681±0.113 | 16.878±0.150 | 20.803±0.173  |
|                      | R2     | 0.945±0.003 | 0.836±0.005  | 0.748±0.008  | 0.631±0.015   |
| AirFormer            | MAE    | 4.881±0.014 | 8.735±0.126  | 11.304±0.032 | 13.380±0.115  |
|                      | RMSE   | 7.400±0.020 | 13.637±0.215 | 16.784±0.045 | 19.951±0.120  |
|                      | R2     | 0.945±0.002 | 0.842±0.002  | 0.755±0.004  | 0.658±0.005   |
| AirPhyNet            | MAE    | 4.995±0.060 | 8.838±0.164  | 11.375±0.152 | 13.900±0.210  |
|                      | RMSE   | 7.416±0.090 | 13.649±0.156 | 16.792±0.214 | 20.708±0.321  |
|                      | R2     | 0.945±0.004 | 0.839±0.005  | 0.754±0.003  | 0.639±0.012   |
| MGSFformer           | MAE    | 4.880±0.015 | 8.850±0.121  | 11.331±0.157 | 14.012±0.352  |
|                      | RMSE   | 7.397±0.013 | 13.683±0.108 | 16.785±0.129 | 20.921±0.225  |
|                      | R2     | 0.946±0.003 | 0.838±0.004  | 0.749±0.011  | 0.635±0.012   |
| Air-DualODE          | MAE    | 4.895±0.018 | 8.835±0.124  | 11.351±0.251 | 13.400±0.156  |
|                      | RMSE   | 7.402±0.020 | 13.640±0.210 | 16.789±0.310 | 19.949±0.133  |
|                      | R2     | 0.945±0.004 | 0.840±0.003  | 0.753±0.004  | 0.659±0.006   |
| Wind-ST-MoE          | MAE    | 4.614±0.019 | 8.387±0.103  | 10.769±0.125 | 12.682±0.170  |

|  |      |             |              |               |              |
|--|------|-------------|--------------|---------------|--------------|
|  | RMSE | 7.181±0.010 | 13.065±0.120 | 16.152 ±0.201 | 19.025±0.210 |
|  | R2   | 0.953±0.002 | 0.849±0.004  | 0.766 ±0.007  | 0.677±0.011  |

Table S4.Comparative experimental results of different methods in the CSJ region.

| Model                | Metric | +1h         | +6h          | +12h         | +24h         |
|----------------------|--------|-------------|--------------|--------------|--------------|
| Persistence          | MAE    | 3.857       | 7.910        | 11.733       | 12.958       |
|                      | RMSE   | 5.560       | 11.777       | 15.290       | 18.862       |
|                      | R2     | 0.937       | 0.738        | 0.440        | 0.280        |
| Seasonal Persistence | MAE    | 3.889       | 6.875        | 11.836       | 11.153       |
|                      | RMSE   | 6.203       | 11.230       | 15.651       | 17.941       |
|                      | R2     | 0.913       | 0.756        | 0.439        | 0.289        |
| ARIMA                | MAE    | 3.759       | 7.895        | 11.726       | 12.054       |
|                      | RMSE   | 5.565       | 11.686       | 15.287       | 17.854       |
|                      | R2     | 0.937       | 0.740        | 0.445        | 0.291        |
| GRU                  | MAE    | 4.545±0.055 | 7.451±0.136  | 10.769±0.244 | 11.542±0.327 |
|                      | RMSE   | 6.451±0.069 | 11.571±0.145 | 14.874±0.207 | 16.554±0.245 |
|                      | R2     | 0.900±0.005 | 0.750±0.010  | 0.511±0.012  | 0.405±0.015  |
| STGCN                | MAE    | 3.762±0.029 | 6.952±0.153  | 10.654±0.246 | 11.242±0.236 |
|                      | RMSE   | 5.452±0.024 | 10.244±0.132 | 14.325±0.320 | 16.451±0.342 |
|                      | R2     | 0.934±0.002 | 0.769±0.007  | 0.561±0.012  | 0.408±0.016  |
| T-GCN                | MAE    | 3.654±0.036 | 6.841±0.142  | 10.545±0.242 | 11.214±0.242 |
|                      | RMSE   | 5.344±0.028 | 10.124±0.112 | 14.254±0.225 | 16.251±0.340 |
|                      | R2     | 0.935±0.003 | 0.774±0.008  | 0.562±0.014  | 0.411±0.015  |
| MTGNN                | MAE    | 3.896±0.048 | 6.763±0.142  | 10.056±0.231 | 12.012±0.257 |
|                      | RMSE   | 5.547±0.033 | 9.921±0.214  | 14.235±0.247 | 16.561±0.243 |
|                      | R2     | 0.932±0.002 | 0.788±0.012  | 0.575±0.013  | 0.402±0.017  |
| HighAir              | MAE    | 3.752±0.023 | 6.828±0.236  | 10.387±0.214 | 11.395±0.221 |
|                      | RMSE   | 5.662±0.029 | 10.521±0.125 | 14.528±0.324 | 16.301±0.244 |
|                      | R2     | 0.928±0.004 | 0.769±0.009  | 0.562±0.019  | 0.410±0.015  |
| GAGNN                | MAE    | 3.721±0.019 | 6.751±0.120  | 10.385±0.205 | 10.974±0.217 |
|                      | RMSE   | 5.621±0.029 | 10.315±0.214 | 14.245±0.214 | 15.963±0.245 |
|                      | R2     | 0.930±0.002 | 0.762±0.011  | 0.568±0.015  | 0.449±0.008  |
| AirFormer            | MAE    | 3.835±0.027 | 6.611±0.128  | 9.891±0.115  | 10.871±0.254 |
|                      | RMSE   | 5.595±0.029 | 10.014±0.123 | 14.051±0.247 | 15.858±0.246 |
|                      | R2     | 0.931±0.003 | 0.774±0.010  | 0.570±0.017  | 0.454±0.011  |
| AirPhyNet            | MAE    | 3.836±0.015 | 6.774±0.242  | 10.722±0.244 | 11.291±0.234 |
|                      | RMSE   | 5.678±0.013 | 10.125±0.125 | 14.908±0.214 | 16.512±0.248 |
|                      | R2     | 0.935±0.002 | 0.772±0.010  | 0.512±0.018  | 0.407±0.012  |
| MGSFformer           | MAE    | 3.756±0.014 | 6.623±0.247  | 9.523±0.254  | 11.254±0.254 |
|                      | RMSE   | 5.550±0.042 | 10.212±0.125 | 13.682±0.175 | 16.504±0.312 |
|                      | R2     | 0.933±0.002 | 0.769±0.008  | 0.584±0.010  | 0.410±0.013  |
| Air-DualODE          | MAE    | 3.825±0.017 | 6.712±0.124  | 9.720±0.158  | 10.952±0.225 |

|             |      |             |              |              |              |
|-------------|------|-------------|--------------|--------------|--------------|
| Wind-ST-MoE | RMSE | 5.663±0.038 | 10.113±0.152 | 13.808±0.157 | 15.988±0.254 |
|             | R2   | 0.932±0.003 | 0.771±0.009  | 0.581±0.020  | 0.448±0.011  |
|             | MAE  | 3.419±0.012 | 6.201±0.112  | 8.641±0.125  | 10.015±0.116 |
|             | RMSE | 5.043±0.029 | 9.360±0.120  | 12.435±0.113 | 14.951±0.204 |
|             | R2   | 0.941±0.002 | 0.802±0.005  | 0.651±0.010  | 0.502±0.011  |

Fig.S1. Sensitivity analysis of different terrain constraint thresholds on the validation set for 24-hour forecasting.

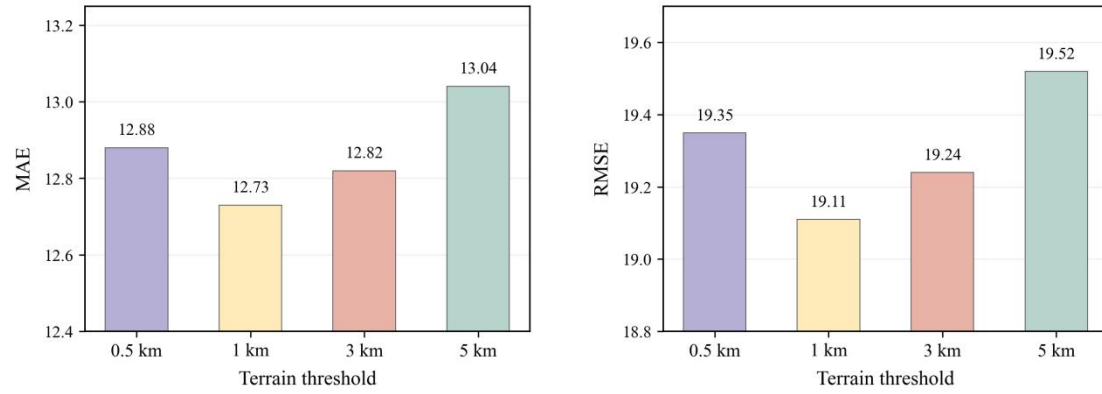


Fig.S2. Distribution of stations in the Beijing-Tianjin-Hebei Region.

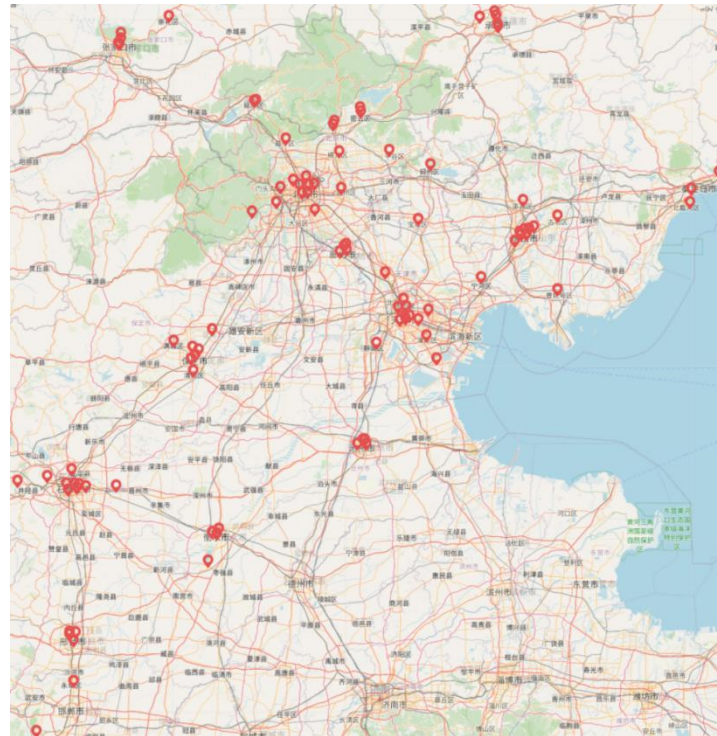
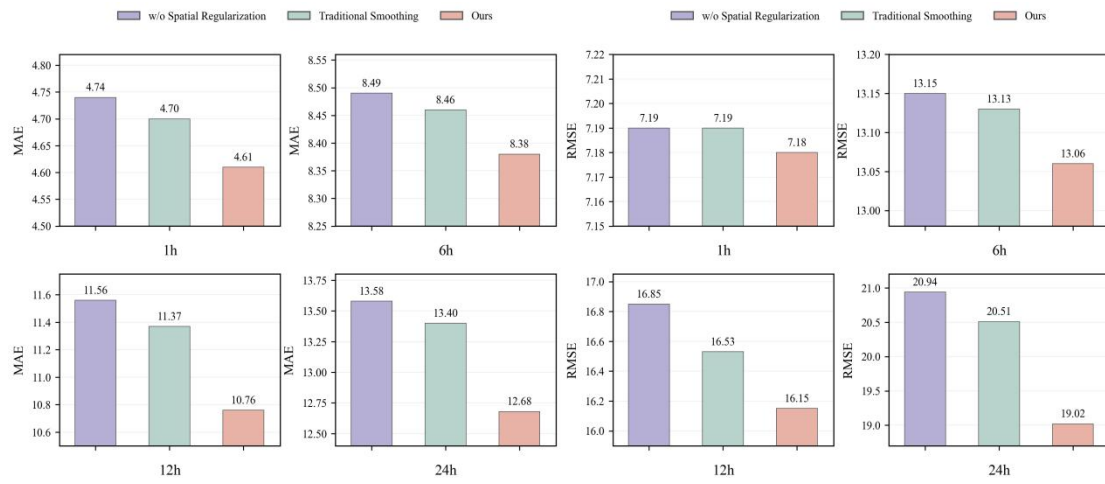


Fig.S3. Comparison of different spatial regularization strategies.



We conduct comparative experiments on different spatial regularisation strategies. Specifically, this paper employs three spatial constraint methods: 1) no spatial regularisation; 2) traditional spatial smoothing methods; and 3) the residual spatial regularisation method proposed in this paper. For short-term forecasting tasks, where recent observations play a dominant role, the performance improvement that spatial regularisation brings is relatively limited. However, as the forecast horizon increases, residual Laplacian smoothing demonstrates a more pronounced advantage, indicating that it is capable of mitigating the cumulative spatial inconsistency errors that arise during long-term forecasting.